

Revisiting the Fundamentals of Swarm Robotics: Event-based Analysis of Biological Swarms

Abstract—Scalability and highly dynamic coordination remain primary challenges in aerial swarm robotics, despite recent advances in the agility of individual autonomous aerial robots. To tackle these challenges, we revisit some of the core methods and assumptions commonly applied in swarm robotics by studying the underlying principles of biological systems that inspire and motivate most of the research in this field. Specifically, we analyze behavior of large-scale flocks of *Columba livia* (rock dove) using data captured by a pair of event cameras. This setup provides novel insights into existing research that relies on standard cameras or GNSS trackers; namely, it allows for microsecond-accurate temporal resolution of fully reconstructed 3D trajectories together with wing motion and other visual cues that may be utilized by the animals for agile coordination. Our analysis suggests that observed biological flocks utilize a *dynamic hierarchy* mechanism, which challenges consensus-based approaches commonly applied in robotics. As we demonstrate by experiments simulating fixed-wing UAVs, a swarming rule based on dynamic hierarchy not only better explains the observed data, but also matches the baselines’ efficiency with more consistent performance. We publish details about our methodology including the design of the event stereo camera device for capturing the data, open-source implementation, as well as the datasets, to stimulate further research of biological swarms and the transfer of these findings to robotic systems.

Index Terms—Biomimetics, Swarm Robotics, Multi-Robot Systems

SUPPLEMENTARY MATERIAL

Additional resources available at <https://bit.ly/4rPcpHV>.

I. INTRODUCTION

The agility of individual Unmanned Aerial Vehicles (UAVs) has rapidly advanced in recent years, reaching a level where they can now rival and even surpass the reflexes of expert human pilots in high-speed navigation [1]. In contrast, multi-UAV coordination lags behind bird flocks capable of highly dynamic maneuvers in cluttered environment while avoiding explicit communication [2]. Despite the much attention bio-inspired approaches have received, understanding of principles underlying their collective behavior remains limited. Our work demonstrates that deeper insights into these mechanisms can mitigate such constraints.

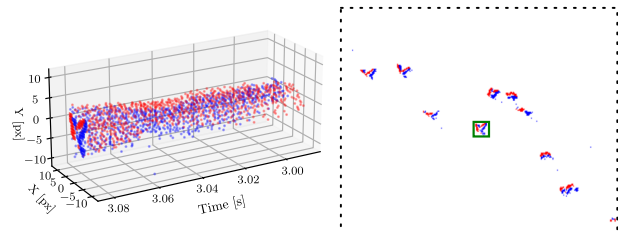


Fig. 1: Microsecond-resolution event tracking of a flock. *Left*: Point-cloud representation of events generated by a single bird over 80 ms (positive events red, negative blue). *Right*: Visualization of the EVENT Frame.

We present a new methodology for studying the collective motion of birds using a stereo event camera (see Fig. 1). Visual event processing enables extracting previously-unavailable information at high temporal resolution: 3D trajectories of individuals within a large flock and visual cues the animals may use for coordination, such as the wingbeat patterns. With this methodology, we have captured several sequences of *Columba livia* flocks (rock dove, also referred to as the common pigeon) during collective maneuvers at an unprecedented scale (the largest analyzed flock has 137 individuals) and temporal resolution (5 ms).

Our analysis of this data shows that the birds follow a flocking rule based on a dynamically switching hierarchy. This observed behavior challenges classical and widely adopted consensus-based swarming algorithms [3]–[5], while supporting more recent approaches centered on dynamic hierarchical reorganization, such as [6], [7]. To evaluate the applicability of these insights to multi-robot systems, we compare four collective behavior models: a consensus baseline [4], a static minimum-time leader [8], PACNav [9], and our proposed dynamic hierarchy model inspired by the observed behavior. The results demonstrate that the dynamic hierarchy model matches the baselines’ efficiency with lower variability of trajectory length and duration.

II. RELATED WORK

Classical consensus-based flocking models are largely influenced by the “boids” model [3], designed to imitate biological swarms visually. It acts on inter-agent relative positions and velocities and assumes omniscient knowledge of agents’ states. We look to biological systems not for aesthetic fidelity but for efficiency: unlike robots, birds have no distance sensors or communication networks, navigating instead on local visual cues [10] and aggregate density rather than discrete individuals [11]. This matters for robotic swarm deployment, where omniscient knowledge cannot be assumed, communication is a bottleneck, and the global post-

hoc statistics typical of the biological literature do not apply, so agents must rely on onboard perception.

A. Hierarchical Structure of Bird Flocks

There are two main approaches to studying bird flocks: using frame-based cameras placed on the ground [10]–[12] and using Global Navigation Satellite System (GNSS) receivers placed on individuals within the flock [13].

The GNSS-based method offers high precision, which revealed several correlations, leading to the proposal of hierarchical leadership as the key principle in inter-flock coordination of pigeons [13]. However, tagging every individual is unscalable for large flocks, and GNSS data is purely metric: it gives absolute position, not visual cues or low-level dynamics that may drive collective behavior.

Multi-camera systems have been used to reconstruct 3D trajectories in massive starling flocks [12], although the analysis presented in the work focused on structural correlations rather than local behavioral mechanisms. Ai et al. [14] propose the “FlockSeer” system for tracking birds using portable smartphone cameras. FlockSeer offers a low-cost, scalable option that is easy to deploy and relies on conventional computer vision methods for the flock analysis.

Event cameras offer a higher dynamic range and temporal resolution in comparison to common frame-based cameras, which enables observing even low-level visual cues. Secondly, only motion is recorded, simplifying the detection of individual birds and reducing data redundancy. Compared to the more established frame-based methods, the challenge is the need for bespoke algorithms for the post-processing, which we tackle in this paper.

B. Bio-inspired Robotic Flocking

Robotic flocking strategies fall into two families, both rooted in principles first observed in animal groups. Reactive, leaderless schemes build on Reynolds’ boids [3], which are modeled on coordinated biological behavior. The best-known real-world demonstration is the evolutionary-optimized, decentralized swarm of 30 outdoor drones by Vászrhelyi et al. [15], whose interaction rules are similarly grounded in statistical-physical models of animal swarms, extending to applications like fixed-wing robots trading communication range for turning rate [5].

The same bio-inspired reactive principle transfers beyond aerial platforms: Berlinger et al. [16] equip an underwater robot collective with purely local, onboard decision rules to reproduce fish-like evasive fountain maneuvers, with no central controller or leader, directly mirroring the predator-evasion behavior of real fish-schools.

Leader-based schemes go the other way, explicitly assigning authority. Mirroring nature, these hierarchies can shift dynamically [17], or remain static by designating a single agent for the entire maneuver [8].

In every case above, the leadership rule is designed or hand-tuned rather than measured. Our model differs here: we derive the leader-election rule from the temporal precedence patterns observed in real pigeon flocks (Sec. V), and use

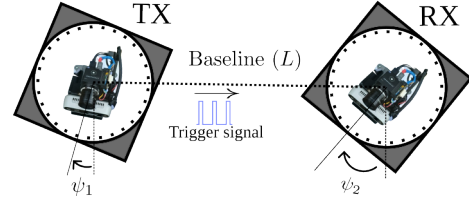


Fig. 2: The proposed device for capturing bird flock data comprising two time synchronized multi-sensor modules.

this rule in a fixed-wing UAV swarm, evaluated against the consensus baseline [4], a static leader [8] and PACNav [9].

C. Contributions

Compared with prior works, we claim the following contributions:

- Design of a stereo event-camera device and an automated pipeline integrating deep learning detection and tracking with multi-view data association for reconstructing 3D trajectories of individuals within high-density flocks.
- A novel dataset of large-scale pigeon flocks performing various maneuvers with a high temporal resolution captured in the wild using the proposed system.
- An analysis showing that pigeons rely on dynamic hierarchy to coordinate movement, and that this, rather than classical consensus, is required to achieve maneuver-independent consistency in decentralized aerial swarms.

We release the data, implementation, and hardware documentation under open source license to facilitate further research in understanding the rules governing biological flocks, which we believe is the key to advancing aerial multi-robot systems.

III. DATA COLLECTION SYSTEM

To capture the relevant biological data, a custom stereoscopic system was designed.

A. Hardware Architecture

The experimental setup consists of two identical multi-sensor modules forming a stereo baseline (see Fig. 2). One module is designated as the *Transmitter*, and the other as the *Receiver*. Each module integrates four primary components:

- 1) **Event camera** (Prophesee EVK4 HD): the primary sensor.
- 2) **RGB camera** (Intel RealSense D435): redundancy and a visual baseline.
- 3) **Inertial Measurement Unit (IMU)** (BMI270): gravity-based tilt alignment.
- 4) **Microcontroller** (STM32U585): time synchronization, IMU readout to PC.

TABLE I: A summary of the captured data sequences.

Date	Spot	Seq. ID	Tot. duration	Num. flights
2025-11-27	1	S1	754 s	4
2026-07-03	2	S2	2348 s	25
	3	S3	457 s	15
	4	S4	462 s	18
	5	S5	1036 s	21

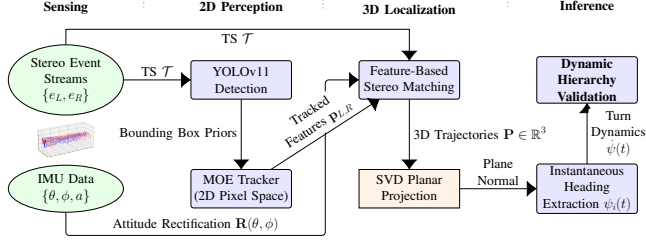


Fig. 3: System architecture for validation of dynamic hierarchy in bird flocks using event-based stereo vision.

For time synchronization, the *Transmitter*'s microcontroller generates a 1 MHz clock that drives the local event camera and is simultaneously propagated via a differential driver over an unshielded twisted pair cable to the *Receiver*'s mother board. Both sensors' internal timestamp counters thus share the same reference with a sub-microsecond precision.

B. Calibration Procedure

Reconstructing complete 3D trajectories of the birds requires precise intrinsic and extrinsic calibration.

1) *Intrinsic Calibration*: We use the standard pinhole model with barrel distortion for all sensors. Since event cameras need relative motion to perceive a calibration pattern, we obtain events triggered by moving edges of squares of a checkerboard, and project latest event polarities to obtain calibration frames, then run standard corner detection and reprojection error optimization to recover focal length (f_x, f_y) , principal point (c_x, c_y) , and distortion coefficients.

2) *Extrinsic Calibration*: The relative pose between the two stereo modules is obtained using physical measurement and inertial sensing:

- **Orientation**: The pitch and roll angles in a gravity-aligned frame are obtained from the gravity force vector measured by the IMUs. The relative yaw is measured with a custom goniometric rig (1° resolution) and a taut line between the modules (see Fig. 2), following [14].
- **Translation**: The baseline distance was manually measured using a tape measure with a centimeter resolution.

C. Dataset Collection

We captured the core dataset over two recording days at five sites (*Spots*) summarized in Table I. Site choice exploited the pigeons' strong site fidelity: the rig captured repeated, dense maneuvers as the flock shuttled between roosts. We segment each recording into flights of 4 s to 45 s, spanning the interval the flock stays within the sensors' Field Of View (FOV), so the flights contain only active flight maneuvers.

Capturing the RGB images negatively affected the stability of the system due to USB bandwidth issues and inflated the recordings with redundant data (roughly $9\times$ the volume compared to only recording event and IMU data, 132 GiB vs. 15 GiB for a single 685 s sequence). Since the RGB images are used for visual qualitative validation, we captured only two recordings with the RGB data included. Events alone give a more stable and scalable system for resource-constrained field platforms.

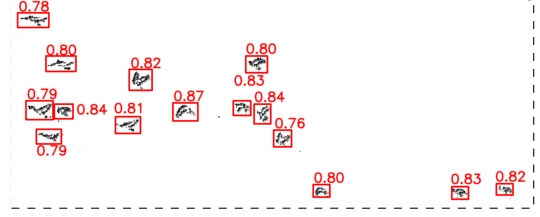


Fig. 4: Bird detections and confidence on the TS, zoomed crop of the region with the highest concentration of birds. Darker color encodes more recent events. White color denotes pixels with no events within the time window.

IV. 3D TRAJECTORY RECONSTRUCTION

This section describes the pipeline for extracting 3D bird trajectories from the stereo event streams (see Fig. 3).

A. Event Representation and Object Detection

Event cameras produce a stream of discrete events $e_k = (x_k, y_k, t_k, p_k)$, where x_k, y_k are the pixel coordinates, t_k is the timestamp and p_k the polarity of the event. To leverage the Convolutional Neural Network (CNN)-based feature extraction, we project this sparse spatiotemporal stream into a dense tensor (i.e., an image) and evaluate two common representations for detection accuracy.

First, the Event Frame (EVF) representation [18] is generated by converting events within a defined temporal window into a 2D image through pixel-wise polarity accumulation. Second, the Time Surface (TS) representation [18] captures the history of the scene by applying an exponential decay kernel to incoming events to create a richer spatio-temporal context. The TS representation is shown in Fig. 4.

The YOLOv11 architecture [19] was trained on a manually labeled subset of the data using a temporal window of $\Delta t = 5$ ms chosen to isolate fast motion and preserve distinct event morphology. We labeled 500 images in total, split 75%/25% into training and test/evaluation sets. As presented in Table II, the TS representation yielded better overall performance, achieving a 1.2 % improvement in mean Average Precision (mAP)@0.5 compared to EVF, with a marginally lower stable-state variance. Based on these results, the TS representation was selected for the trajectory extraction pipeline.

B. Multi-Object Event Tracking

We have observed that standard visual tracking algorithms struggle with visual similarity, high density, small objects, and frequent occlusion present in our data. Our custom Multi-Object Event (MOE) tracker addresses this in two stages: an online pass produces intermittent but consistent object tracks, which an offline pass re-links across gaps using the birds' wingbeat signatures as a matching feature.

Let $\mathcal{D}_k = \{\mathbf{d}_k^1, \dots, \mathbf{d}_k^M\}$ be the detections from the YOLOv11 model at time step k , each $\mathbf{d}_k^j = [x \ y]$ the centroid of the j -th bird, and $\mathcal{T}_{k-1} = \{T_{k-1}^1, \dots, T_{k-1}^N\}$ the

TABLE II: Performance with different event representations.

Method	Prec.	Recall	mAP@0.5	mAP@0.5 var. [†]
EVF	0.9064	0.88755	0.93521	0.0023
TS	0.9270	0.90745	0.94737	0.0044

[†] Variance over the last 10 training epochs.

set of active tracks at time step $k - 1$, the i -th with state $S_k^i = \{\mathbf{p}_k^i, \mathbf{v}_k^i\}$ (position, velocity).

1) *Online Association*: Before new detections arrive at the time step k , the position of each track is predicted using a constant-velocity model as $\hat{\mathbf{p}}_k^i = \mathbf{p}_{k-1}^i + \Delta t \mathbf{v}_{k-1}^i$, $\Delta t = t_k - t_{k-1} = 5$ ms. Detections are divided by confidence into a high set ($\text{conf} \geq \tau_h = 0.5$) and a low set ($0.1 \leq \text{conf} < \tau_h$), the rest being discarded. Following a ByteTrack-style two-round scheme [20], high-confidence detections are matched first and may initiate or continue tracks, while low-confidence detections may only continue still-unassigned tracks. Recovering the weak detections that YOLOv11 emits under motion blur and overlap markedly reduces fragmentation without admitting spurious tracks.

Each round solves the assignment greedily using the cost $C_{ij,\text{dist}} = \|\mathbf{d}_k^j - \hat{\mathbf{p}}_k^i\|$, gated by $\|\mathbf{d}_k^j - \mathbf{p}_{k-1}^i\| < \min(\lambda_v \tilde{v}, h_{\max})$, where $\tilde{v}$ is the per-flight median speed (robust to the wide speed spread within a maneuver), λ_v a tolerance coefficient, and $h_{\max}$ a hard displacement cap. On a successful match to detection j^* , the position is updated from the measurement and the velocity smoothed by an exponential filter as $\mathbf{v}_k^i = \alpha_v \tilde{\mathbf{v}}_k^i + (1 - \alpha_v) \mathbf{v}_{k-1}^i$, where α_v is a smoothing coefficient, and $\tilde{\mathbf{v}}_k^i = \frac{\mathbf{d}_k^{j^*} - \mathbf{p}_{k-1}^i}{\Delta t}$.

2) *Offline Track Fragment Linking*: Dense flocks cause frequent identity ambiguities, so the online pass stays deliberately conservative, ending a track after a short unassociated span, which results in track fragments that are then re-linked in a second round. For every fragment pair separated by a gap of at most $g_{\max} = 15$ frames, the last state of the first is integrated forward and the first state of the other backward. A seam cost $C_{ij,\text{seam}}$ is defined as the residual position error, and links whose seam cost exceeds $\min(\lambda_g g, h_{\max})$, where λ_g is a tolerance coefficient, and g is the gap, are discarded.

The stream’s high temporal resolution exposes a cue frame-based trackers lack: the periodic wingbeat. Per fragment, we extract the dominant frequency f_w via Fast Fourier Transform of the flapping-induced vertical velocity within a plausible 2–12 Hz band. The total matching cost is $C_{ij,\text{match}} = C_{ij,\text{seam}} + w_w |f_{w,i} - f_{w,j}|$, where w_w is a weighting coefficient. Fragments are therefore joined based on both the trajectory and flapping signature, helping to resolve crossings that are ambiguous under position alone.

To suppress transient mismatches, the online association is run forward and backward in time, and only links confirmed in *both* passes are retained before bridging. Links present

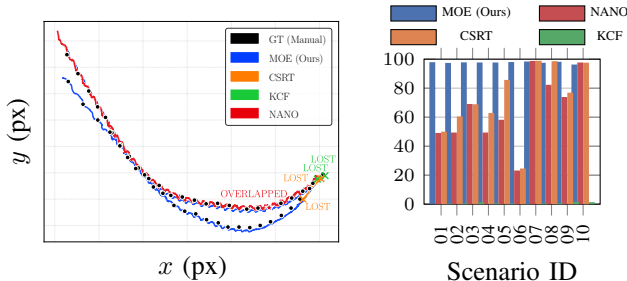


Fig. 5: Experimental evaluation of different tracking systems. *Left*: A qualitative comparison on S1-011. *Right*: SR (%) across 10 scenarios.

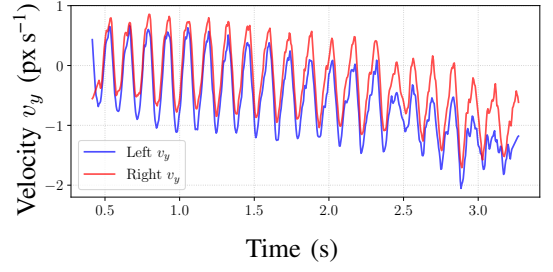


Fig. 6: Vertical velocities over time for a single bird in the left and right images. The velocities are correlated due to the wing flapping ($\rho_v = 0.97$). In a single pass, typically a momentary occlusion or false positive, are dropped.

3) *Evaluation*: The tracker’s parameters ($\lambda_v, h_{\max}, \alpha_v, \lambda_g, g_{\max}, w_w$) are optimized with Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [21] against 28 manually annotated reference tracks (2 birds within a flock of 20 from the flight S1-011, 5.6 s long, and the complete 26 bird-flock annotated in flight S1-021, 6.4 s long).

The maximized criterion was the Success Rate (SR), defined as

$$\text{SR} = \frac{1}{N} \sum_{t=1}^N \mathbb{1}(E_t < \gamma) \times 100\%, \quad (1)$$

where $\mathbb{1}(\cdot)$ is the indicator function, E_t is the location error w.r.t. the manual annotation, and $\gamma = 50$ px is an error threshold. The optimized configuration reaches $\text{SR} = 88.5\%$.

We evaluate the MOE with CSRT [22], KCF [23], and NanoTracker¹ on a set of 10 stress-testing simulated benchmarking sequences comprising 40 trajectories, and on the two manually annotated tracks. On the annotated tracks, where the birds initiate flight in close proximity, NanoTracker merges the trajectories and the other trackers fail (Fig. 5-left), while MOE tracks correctly. The stress-testing trajectories confirm the same pattern (Fig. 5-right, and the attached video): MOE maintains accurate tracking where the baselines fail. We attribute this to the visual similarity of the birds and occlusions, which MOE can overcome thanks to the offline merging and the wingbeat cues.

C. Stereo Matching and 3D Trajectories

To reconstruct the 3D trajectories from the sparse event streams, we initially attempted to employ an established event-based stereo method, the DDES [24], which proved unsuitable for our dense flocks of birds and their fast, overlapping motion. We instead developed a specialized approach that turns 2D pixel-space tracks in the left and right image into metric 3D coordinates via a multi-feature association cost and linear trilateration.

1) *Image Rectification and Feature Extraction*: Given the intrinsic and the extrinsic parameters obtained using the calibration procedure described in section III-B.2, the rectification transform and projection matrix are computed for each camera. Each track centroid $\mathbf{d} = [x \ y]$ is transformed into the rectified space $\mathbf{d}' = [x' \ y']$ such that the epipolar lines

¹<https://github.com/HonglinChu/NanoTrack>

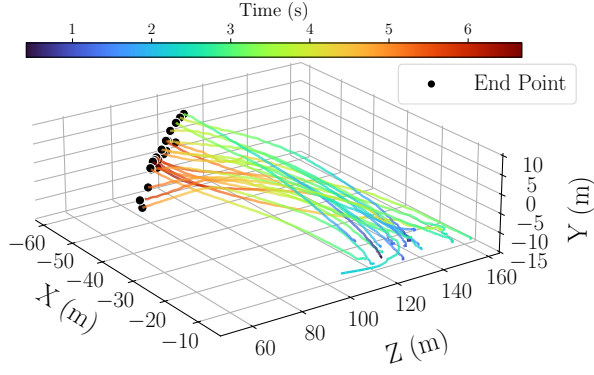


Fig. 7: Reconstructed 3D trajectories of the bird flock from flight S1-006. are horizontally aligned, satisfying the constraint $y'_L \approx y'_R$, i.e. the y -coordinate of an object is almost equivalent in the left and right images. A Savitzky-Golay filter with window length $n = 21$ and order 2 is applied to the rectified coordinates to produce smoothed trajectories. This provides a cleaner signal for further correlation analysis.

2) *Multi-Feature Association*: The correspondence problem is formulated as a global optimization task. For every pair of tracked trajectories from the left and right cameras that share a minimal temporal overlap, a cost function C_{total} is defined based on three constraints:

$$C_{\text{total}} = w_y \cdot \bar{E}_y + w_m \cdot (1 - \rho_v) + w_s \cdot (1 - \rho_y), \quad (2)$$

where $\bar{E}_y$ is the mean vertical epipolar error, $\rho_v, \rho_y \in [0, 1]$ represent correlation coefficients of the vertical velocities and trajectory shapes, and w_y, w_m, w_s are weight parameters.

As in Sec. IV-B.2, the wingbeat-induced movement ensures that the motion profiles of the same individual are highly correlated in the left and right images, as shown in Fig. 6. The optimal set of matches is found using the Hungarian algorithm by leveraging these unique motion signatures to resolve ambiguities even in dense flocks.

To select the weights w_y, w_m, w_s , a ground truth dataset was established by manually matching 28 trajectories across the stereo views based on visual inspection. The weight parameters were then tuned to maximize the assignment accuracy against this baseline, which yielded the final values of $w_y = 2$, $w_m = 40$, and $w_s = 20$ with a 91.6% accuracy.

3) *Triangulation*: For each matched pair of rectified centroids $\mathbf{d}'_L = [x'_L \ y'_L]$ and $\mathbf{d}'_R = [x'_R \ y'_R]$, the 3D position is calculated via linear triangulation. We solve the overdetermined system of equations $\mathbf{A}\hat{\mathbf{x}} = \mathbf{0}$ for $\hat{\mathbf{x}}$, where the matrix $\mathbf{A}$ is constructed from the left and right projection matrices $\mathbf{P}_L, \mathbf{P}_R$ as

$$\mathbf{A} = \begin{bmatrix} x'_L \mathbf{P}_L^{3\tau} - \mathbf{P}_L^{1\tau} \\ y'_L \mathbf{P}_L^{3\tau} - \mathbf{P}_L^{2\tau} \\ x'_R \mathbf{P}_R^{3\tau} - \mathbf{P}_R^{1\tau} \\ y'_R \mathbf{P}_R^{3\tau} - \mathbf{P}_R^{2\tau} \end{bmatrix}, \quad (3)$$

where $\mathbf{p}_i^n$ denotes the n -th row of the projection matrix i . The resulting 4D homogeneous vector $\hat{\mathbf{x}}$ is normalized to yield the metric coordinates $\mathbf{x} = [X \ Y \ Z]$ in the reference camera frame. This pipeline enables the resolution of complex, intertwining flight paths within high-density flocks, as shown in the 3D reconstruction in Fig. 7.

V. DYNAMIC HIERARCHY IN PIGEON FLOCKS

To investigate properties of navigational authority, we analyze the temporal structure of flight via time-delayed directional correlation, following Nagy et al. [13].

A. Dimensionality Reduction and Heading Extraction

Analysis of the trajectories reveals that flight maneuvers are mostly spatially concentrated to a local plane specific to a concrete maneuver. This observation is directly supported by previous studies on large-scale starling flocks, which also reports that collective turns predominantly occur within a local 2D plane [12]. Consequently, Singular Value Decomposition (SVD) was applied per-maneuver to the reconstructed 3D trajectories to project the motion onto a plane defined by the first two singular vectors $\mathbf{u}$, and $\mathbf{v}$. Within this local plane, the instantaneous heading for the i -th agent at time step k is computed as $\psi_k^i = \arctan 2(\dot{v}_k^i, \dot{u}_k^i)$, and its unit heading vector as $\mathbf{h}_k^i = \frac{\bar{\mathbf{v}}_k^i}{\|\bar{\mathbf{v}}_k^i\|}$, where $\bar{\mathbf{v}}_k^i = [\dot{u}_k^i, \dot{v}_k^i]$ is the agent's velocity in the plane $\mathbf{u}, \mathbf{v}$, obtained through differentiation of its projected position $\bar{\mathbf{x}}_k^i$. The normalized heading is utilized to isolate the directional intent from individual flight speeds, ensuring that correlation metrics are not skewed by velocity magnitudes.

B. Time-Delayed Directional Correlation

Hierarchy is measured throughout the maneuver as *temporal precedence* in heading. For an ordered pair of agent indices (i, j) , we define the time-delayed directional correlation $C_{ij}(\tau)$, where τ is the delay in time steps, as the mean cosine similarity of the two vectors:

$$C_{ij}(\tau) = \langle \mathbf{h}_k^i \cdot \mathbf{h}_{k+\tau}^j \rangle_k. \quad (4)$$

The *characteristic delay* of the pair is the maximizer $\tau_{ij}^* = \arg \max_{\tau \in [-\tau_{\max}, \tau_{\max}]} C_{ij}(\tau)$. A positive maximizer $\tau^* > 0$ translates to agent j following agent i with delay τ^* , consequently $\tau^* \approx 0$ denotes synchronous turning. The i -th agent's *leadership score* $\mathcal{L}_i$ is then defined as

$$\mathcal{L}_i = \frac{1}{|\mathcal{P}_i|} \sum_{j \in \mathcal{P}_i} \tau_{ij}^*, \quad (5)$$

where $\mathcal{P}_i = \{j \mid C_{ij}(\tau_{ij}^*) \geq 0.2\}$.

To decide whether leadership is *static* or *dynamic*, $\mathcal{L}_i$ is recomputed locally in a sliding window of length $W = 1.5$ s advanced in steps of 5 ms. The leader is identified in each window as the bird with the highest score. The leadership is dynamic when more than one distinct leader exists during the maneuver. Finally, the leader-follower relations $i \rightarrow j$ with characteristic delays τ_{ij}^* form a directed leadership graph. The linearity of this structure is quantified by the triangle transitivity, t_{tri} [13]. For fully connected triads of agents, t_{tri} computes the proportion of triads that exhibit a linear, transitive order rather than a cyclic one. It is rescaled so that $t_{\text{tri}} = 0$ matches the transitivity expected under random pairwise relations. While t_{tri} is scalar, it evaluates local topological interactions, capturing the presence of a multi-level hierarchy that a 1D averaged ranking would obscure.

TABLE III: Evidence of dynamic hierarchy for 10 representative flights.

Flight	Num. birds	Num. chg.	Lvl.	t_{tri}	Edge (R)	f_w (Hz)
S1-006	19	3	10	0.95	0.000	3.65
S1-021	26	3	5	0.79	0.032	3.64
S2-004	58	6	7	0.83	0.024	3.75
S2-007	103	5	3	0.91	0.254	5.22
S2-010	137	4	5	0.73	0.090	3.96
S2-014	23	2	15	1.00	0.109	3.38
S3-004	10	2	8	1.00	0.002	3.53
S4-017	5	1	4	1.00	0.000	4.09
S5-001	60	1	5	0.75	0.000	4.15
S5-017	48	2	21	1.00	0.068	4.10
Median (20 flights)	30	3	8	0.96	0.020	3.79

Num. chg.: leader changes, *Lvl.*: hierarchy levels, t_{tri} : triangle transitivity of the leadership network, *Edge*: distance from the temporal leader to the flock convex hull normalized w.r.t. the mean flock radius R , f_w : median wingbeat frequency.

C. Results: Asynchronous Propagation

In the collected data, we selected 20 flights that contain a turning maneuver (the rest consist of straight trajectories, takeoffs, and landings, where the leadership cannot be easily determined or remains static). Table III summarizes the results and Fig. 8 illustrates a representative maneuver.

The characteristic delays form a staggered cascade rather than a simultaneous response, suggesting that the flock reacts to localized visual cues rather than to a global metric average. The leadership graphs are strongly transitive (median $t_{tri} = 0.96$), in agreement with the hierarchies reported by Nagy et al. [13]. To quantify this, we analyzed the nodes in the directed leadership graphs across all flights, identifying the intermediate relay birds. These are nodes possessing an in-degree ≥ 1 (they turn after a predecessor) and an out-degree ≥ 1 (they turn before their follower), revealing that 46.4% of birds simultaneously lead some flock mates and follow others, which implies a coexistence of hierarchy and role-mixing. The median of the distance from the leader to the convex-hull's edge relative to the radius of the flock is 0.02 meaning that leaders tend to be near the flock's boundary.

We summarize these measurements into two main observations that indicate what a robotic analogue of this behavior should look like:

- O1 Leadership propagates sequentially through local reactions to immediate predecessors, with individuals frequently acting as intermediate relays. Thus, any robotic election rule should rely on local, perception-based interactions to form a dynamic hierarchy.
- O2 The main leader typically resides near the flock's edge, suggesting this individual is closest to the general goal.

VI. APPLICATION TO FIXED-WING ROBOTIC SWARMS

To evaluate these biological principles on aerial robotic swarm, we designed a comparative simulation of fixed-wing UAV platforms, whose non-holonomic dynamics resemble those of birds. We use a high-fidelity UAV simulator [25] and extend it with the fixed-wing model derived by Song et al. [26] (namely eq. (1)) to describe agents' dynamics. The authors of [26] validated the model, with data obtained in the real-world flight with 18 UAVs. The FOV of each agent is restricted to 316.4° according to the biological counterpart [27] and the agents react to their neighbors within

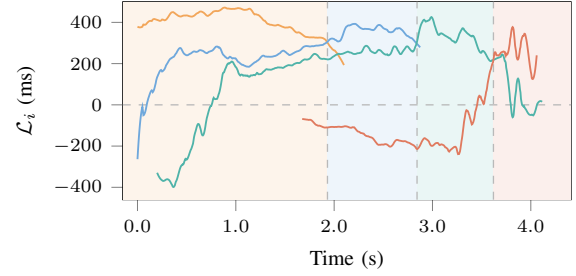


Fig. 8: Leadership score $\mathcal{L}_i$ of the leader individuals over time during flight S1-006 (19 birds). Each line represents one bird, the background color marks who leads in each interval, and dashed lines mark the 3 leader changes.

their FOV only. We compare four swarming paradigms: a global consensus [4], a hierarchical static-leader election method [8], a bio-inspired collective navigation approach with a dynamic leader [9], and our proposed dynamic hierarchical navigation based on presented biological findings.

A. Global Consensus Baseline

As the consensus representative, we adopt the flocking algorithm of Olfati-Saber [4], the canonical formalization of Reynolds' rules [3] with formally provable aggregation and velocity-matching properties. In this model, no agent is privileged, all agents follow the same navigational rule based on perception of their neighbors and the common goal.

B. Minimum Time-Consensus Static Hierarchy

As a static-leader representative, we follow the leader selection method from [8]. In the method, the flock is abstracted as single integrators under the consensus law from [4], where only the leader knows the goal and steers toward it. Given an initial configuration and a Laplacian of the swarm's communication graph, the leader is selected as the agent that minimizes the consensus time over possible inputs, which is solved using a Gröbner basis-based state-space partitioning. For more details, we refer the reader to the original publication [8].

The optimality guarantee holds only under the assumption of linear consensus dynamics. However, this assumption is violated by the nonlinear, FOV-restricted dynamics of the simulated model. We nonetheless adopt the criterion as a principled way to fix the initial hierarchy.

C. PACNav

We further include PACNav [9], a bio-inspired collective navigation approach for swarms deprived of communication and external localization. The work assumes an informed agent that navigates towards the goal according to designed rules, while the rest pursue it under a perception-based collision-avoiding rule. We modify the algorithm to elect the informed agent at every waypoint using the election signal defined as persistence criterion specific to the algorithm [9]. As a result, the leading authority is being passed and the work provides a dynamic-leadership baseline.

D. Dynamic Hierarchy

We design the flocking rule on the basis of the two main observations stated in Sec. V-C. The proposed model utilizes the same follower law as methods described in VI-A and VI-B defined by eq. (8). However, instead of designating the leader once, it is re-elected *locally* and *continuously* (unlike VI-C), following O1.

The election protocol is based on metric inspired by observation O2 defined as:

$$s_i = \exp\left(-\frac{\delta_i^2}{2\sigma^2}\right), \quad \delta_i = \min_k \text{dist}(\mathbf{p}_i, \overline{\mathbf{q}_k \mathbf{q}_{k+1}}), \quad (6)$$

where δ_i is the distance from the i th agent's current position $\mathbf{p}_i$ to the nearest segment $\overline{\mathbf{q}_k \mathbf{q}_{k+1}}$ of the reference path and σ is a parameter determining how sharply the score decays with the distance δ_i . The metric described by eq. (6) is calculated in a decentralized manner over each agent's local election neighborhood $\mathcal{N}_i^{\text{ldr}}$ restricted by the election radius r_{ldr} and its FOV. Agent i considers itself a local leader ($\ell_i = 1$) if no neighbor exceeds its s_i score:

$$\ell_i = \begin{cases} 1, & s_i \geq \max_{j \in \mathcal{N}_i^{\text{ldr}}} s_j, \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

A local leader ($\ell_i = 1$) tracks the path attraction term and ignores the relay term of eq. (8), while a follower ($\ell_i = 0$) pursues the higher-scoring neighbor that becomes its relay in eq. (8). Importantly, an ego-agent classified as follower by eq. (7) can simultaneously be a maximizer of a different neighborhood $\mathcal{N}_j^{\text{ldr}}$ representing the j th agent's perspective. This mechanism naturally mimics the role-mixing behavior observed in the biological data. The hierarchy dynamically shifts as the flock navigates: whenever changing geometry causes a follower's score to surpass its current relay, it automatically takes the lead. Furthermore, due to the proposed election protocol reflecting the active FOV constraints, restricted vision is not a vulnerability; if a leader turns out of sight, the follower seamlessly elects a new one.

Leadership changes are therefore not triggered by an artificial timer or a hand-set hysteresis but by the geometry itself. As the flock advances and the reference path bends, the agent closest to the upcoming segment changes. Due to distributed, per-agent evaluation of eq. (7), the leader-follower structure emerges as a complex layered graph, reproducing the coexistence of hierarchy and role-mixing measured in Sec. V-C.

The external input guides the global leader toward the goal, while every other agent follows a *relay*, the neighbor with the lowest relay count to the leader. To avoid collisions, a pairwise repulsion term $\mathbf{v}_{\text{rep}}$ over neighbors within the radius r_{rep} was added. The desired velocity $\mathbf{v}_{i,\text{des}}$ is governed by a unified control law that, depending on the agent's leadership status (ℓ_i), shifts between pursuing the local relay and tracking the reference path. Because the topology varies during a maneuver, a relay may leave a follower's restricted FOV; the leadership status then lets the agent fall back on the path term:

TABLE IV: Comparison of the navigation models over the 20 flights.

Method	Traj. length $\bar{L}$ (m)	Duration t_{traj} (s)	Cohesion $\bar{d}_{\text{coh}}$ (m)
Global Consensus [4]	843 $\pm$ 491	26.4 $\pm$ 14.0	36.8 $\pm$ 8.1
Static Hierarchy [8]	997 $\pm$ 502	31.0 $\pm$ 14.4	72.7 $\pm$ 22.1
PACNav [9]	829 $\pm$ 503	28.9 $\pm$ 14.5	62.0 $\pm$ 19.2
Dynamic Hierarchy (ours)	834 $\pm$ 255	26.1 $\pm$ 7.3	54.3 $\pm$ 17.3

Length, duration, and cohesion are mean $\pm$ std. dev. across flights. Best in bold.

$$\mathbf{v}_{i,\text{des}} = w_{\text{rep}} \mathbf{v}_{\text{rep}} + \ell_i w_{\text{path}} \mathbf{v}_{\text{path}} + (1 - \ell_i) \left(w_{v,\text{rly}} \mathbf{v}_{\text{rly}} + w_{p,\text{rly}} \frac{\mathbf{p}_{\text{rly}} - \mathbf{p}_i}{\|\mathbf{p}_{\text{rly}} - \mathbf{p}_i\|} \right), \quad (8)$$

where $\mathbf{v}_{\text{path}}$ is the path attraction term, and w_{rep} , $w_{v,\text{rly}}$, $w_{p,\text{rly}}$, and w_{path} are weighting coefficients. To ensure a fair evaluation that isolates the effect of the topology from the low-level flight dynamics, all three hierarchical models, the Static Hierarchy [8], PACNav [9] and our Dynamic Hierarchy, execute this exact control law. Only the leader election mechanism, which dictates how the leadership status ℓ_i is assigned, differs among these hierarchical models.

E. Experimental Protocol

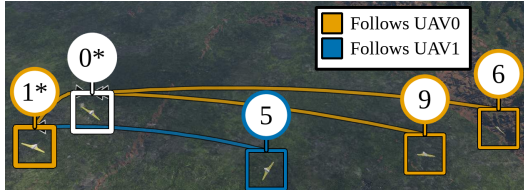
The number of agents, their initial positions, velocities, and target waypoints are extracted from the empirical data of the 20 analyzed flights described by Table III. Since the stereo tracks are fragmented, each flight uses the longest window over which a fixed set of birds is tracked throughout. The reference path is the center-of-mass trajectory of the observed flock.

To avoid tuning bias, all controllers were optimized independently using CMA-ES [21] over their specific parameters (interaction/repulsion radii and weights $w_{(\cdot)}$ for the leadership models, spacing d and gains $c_{(\cdot)}^\alpha, c_{(\cdot)}^\gamma$ for Olfati-Saber). The optimization was performed over the 20 empirical flights, applying the same disturbance model (neighbor-state jitter and wind-like velocity noise) used for the final evaluation. To ensure robust performance, each candidate is evaluated over a Monte-Carlo ensemble of these disturbances, minimizing the penalized cost $\mathbb{E}[J] + \lambda_\sigma \text{std}[J]$ ($\lambda_\sigma = 30$), where the base fitness J is:

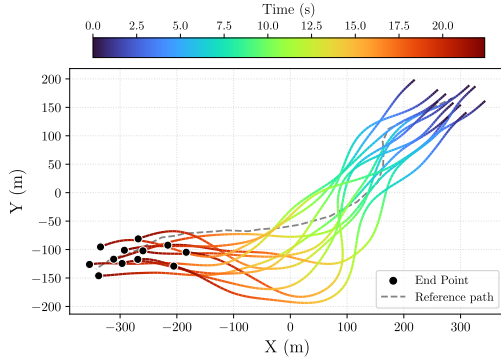
$$J = \lambda_1 \bar{e}_{\text{pos}} + \lambda_2 \bar{a} + \lambda_3 \bar{d}_{\text{coh}} + \lambda_4 t_{\text{traj}} + p_{\text{coll}}, \quad (9)$$

$\bar{e}_{\text{pos}}$ is the mean distance of the agents to the reference path, $\bar{a}$ the mean acceleration magnitude (reflecting lateral control effort during turns, as airspeed is constant), $\bar{d}_{\text{coh}}$ the mean spatial dispersion, t_{traj} the time to reach the goal, and p_{coll} a penalty ($= 1000$) for breaching the minimum inter-agent distance. The coefficients $\lambda_1 = 15$, $\lambda_2 = 5$, $\lambda_3 = 2$, $\lambda_4 = 0.1$ are set empirically to prioritize tracking and smoothness while maintaining compactness and time efficiency. Throughout, we refer to trajectory length $\bar{L}$ and duration t_{traj} jointly as *efficiency*.

Table IV and Fig. 9 summarize the results under perception and wind-like noise. Dynamic Hierarchy matches the mean efficiency of Global Consensus while being far more robust across maneuvers: an F-test ($\text{df} = 19, 19$) shows significantly lower variance in both trajectory length and duration against Global Consensus ($F = 3.70$, $p = 0.0065$; $F = 3.69$,



(a) A visualization of an experiment in the simulation environment [25]. Bounding boxes are colored by each UAV's current relay, with an arrow pointing to it; asterisks mark the local leaders ($\ell_i = 1$).



(b) Trajectories of the UAVs during the maneuver, top-down view.

Fig. 9: Dynamic Hierarchy flocking model executing the recorded trajectory from flight S1-006 in a simulated experiment.

$p = 0.0066$), Static Hierarchy ($F = 3.87$, $p = 0.0050$; $F = 3.90$, $p = 0.0047$), and PACNav ($F = 3.88$, $p = 0.0049$; $F = 3.94$, $p = 0.0044$). This robustness follows from re-electing the leader by local geometry, which lets the swarm adapt to path changes as they happen. Global Consensus reaches tighter mean cohesion, but the number misleads: it comes from an unnaturally rigid synchronization that fails badly on sharp turns. In the sharpest turn (corresponding to S5-008) of the set, its fixed heading update produced a path three times longer than usual (2935 m vs. 700–900 m), an overshoot Dynamic Hierarchy avoided outright. Static Hierarchy was the least efficient of the four, confirming that a swarm's topology must adapt as the maneuver unfolds.

VII. CONCLUSION

This work advances the foundations of swarm robotics by investigating the interaction mechanisms governing collective behavior in biological flocks. We introduced a stereo event-camera system and an automated pipeline for extracting 3D trajectories of individuals within large-scale bird flocks. The proposed system was used to collect a dataset of pigeon flocks with actuation-level visual detail and unprecedented temporal resolution. Analysis of multiple flock maneuvers supports the hypothesis of dynamically evolving navigation hierarchy, which we translated into a flocking strategy for multi-robot systems. Using local-perception leader election, the proposed strategy matched the efficiency of a consensus baseline in trajectory length and completion time, delivered the most consistent behavior, and outperformed a fixed leader, highlighting the practical value of biologically grounded interaction models. The released dataset and methodology open further avenues for studying

collective behavior and its transfer to robotic swarms. Validating the proposed dynamic hierarchy in real-world flight is the natural next step toward physical swarm deployment.

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